

An Investigation into How Autonomous Vehicles Change Travel And Activity Behavior Among Transport Disadvantaged Persons

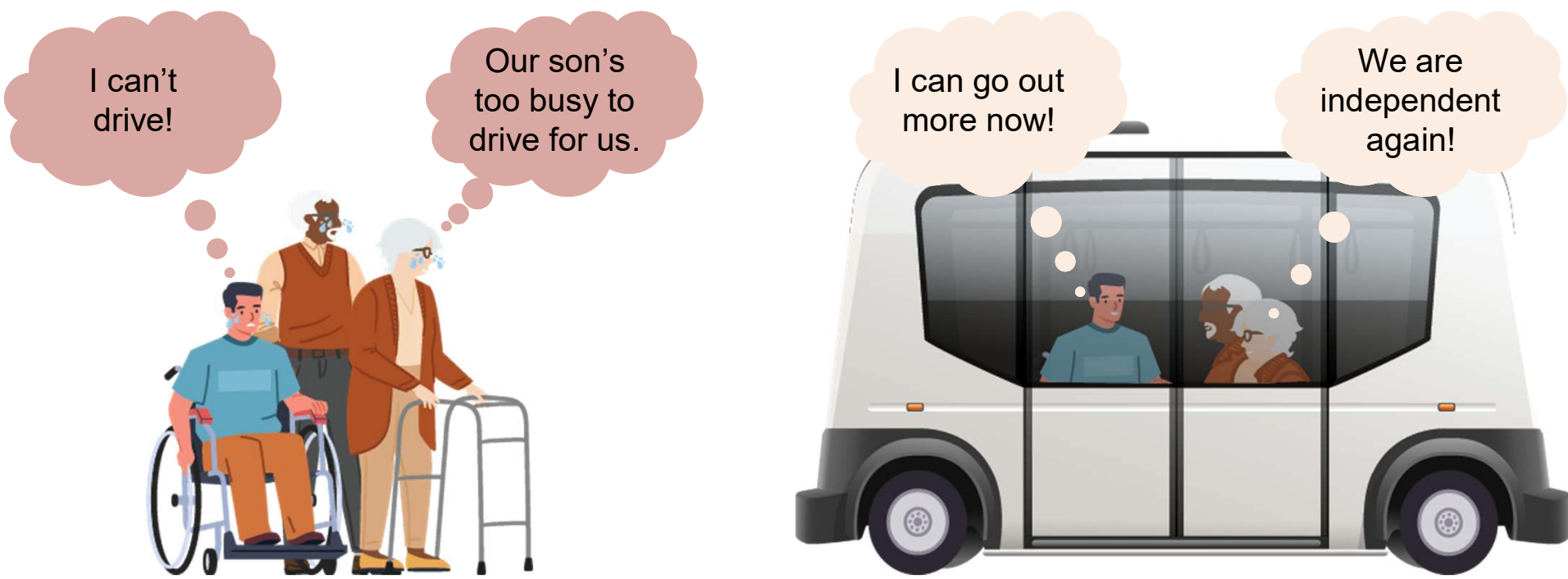
Arkar Than Win, John D. Nelson, Chinh Ho, Institute of Transport and Logistics Studies, The University of Sydney

Introduction

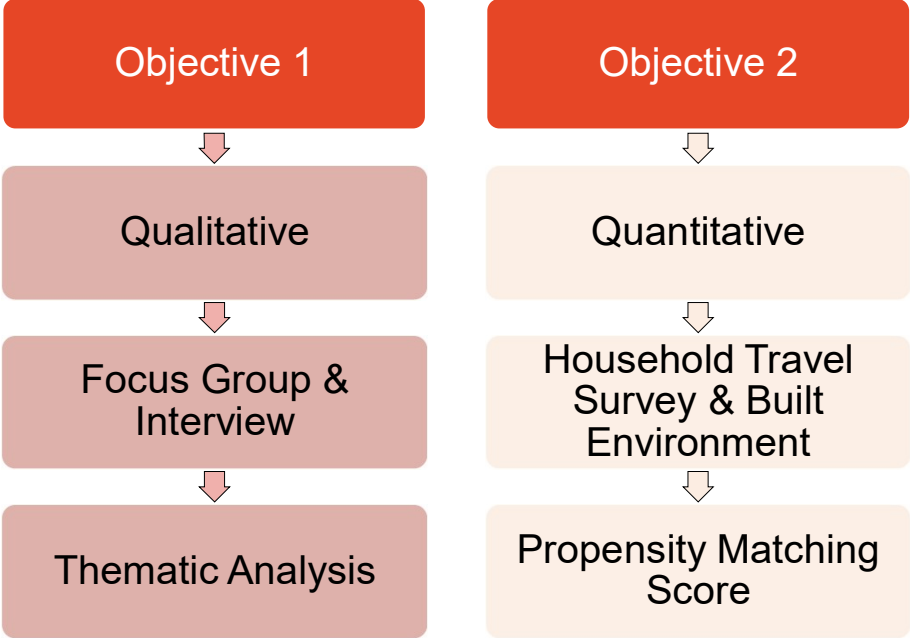
- Transport is essential for both seniors and people with disabilities (PWD), as it enables access to basic needs, healthcare, and social participation. However, their travel needs often remain unmet, partly due to limited access to transport options.
- Level 5 autonomous vehicles (AVs), capable of fully self-driving without human input, may enhance mobility for these transport-disadvantaged groups.

Objectives

- Explore the potential of AVs to improve mobility for seniors, PWD, and other non-drivers using Greater Sydney Metropolitan Area as a case study.
- Predict changes in travel and activity frequency resulting from AV access and use.



Research Design

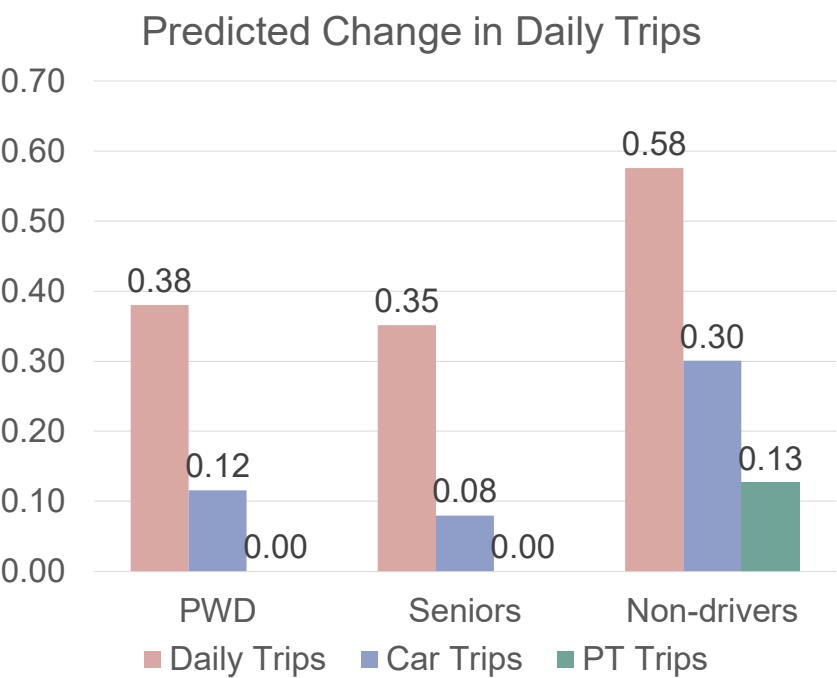


Preliminary Findings: Quantitative Research

After controlling for socio-economic, demographic, household, and built environment factors,

- PWD and seniors make 0.4 fewer trips per day compared to those without disabilities and younger adults.
- Non-drivers make 0.6 fewer trips per day compared to drivers.

Assuming AVs address mobility barriers related to disability, ageing, and not being able to drive, **the potential increase in daily trips** enabled by AVs among these groups is shown below.



Preliminary Findings: Qualitative Research

- Most seniors, and PWD view AVs as beneficial for **overcoming physical limitations**, **reducing reliance on others**, and increasing their ability to **make more trips**.
- Some participants expressed concerns about the **loss of human assistance**.

"If I have the chance, I would use AVs, because it would be freedom for me, so I could just go"

"100% proven and safe, I would use AVs. If I had one of these at my disposal, sky is the limit. I could go anywhere. Brisbane, Darwin, or the beach"

- AVs are also seen as helpful in reducing risks from **reckless drivers and scammers**.

"Taxi drivers wanted me to do a special deal with him. He said, 'Don't use that card as it's rubbish'. This is the taxi card, issued by the government."

The bus driver went around the corner, he went too fast...that's why I fell. I hurt my back...since then, I had a bad back."

"Because of the medication I'm on, I'll be falling asleep and need someone to wake me up"

"It is nice to know that there's somebody watching you and assisting you...if you're going up...very difficult for people on walker...might lose your balance"

- Some fear AVs might reduce travel enjoyment and **eliminate valued social connections** with drivers.

"Get into AVs, not talk to anyone...because no one there... go home and not talk to anyone there, and that sort of interaction that put you back in the community"

- Some expressed concern over limited tech skills and the challenge of **communicating their needs** to AVs.

"I prefer someone that understand my needs. How would you direct AVs where to go when it arrive at your place? How to make them come in the driveway?"

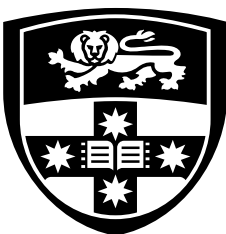
Participants' Information

- 2 focus groups & 3 in-depth interviews
- 4 PWDs & 7 seniors of whom 4 have disabilities
- All live in New South Wales

Conclusions

AVs present a **double-edged sword** – they can enhance mobility and independence for seniors and PWD but must be implemented with care. AVs should:

- Compensate for the loss of human assistance
- Be designed to understand and respond to a diverse range of needs.
- Be prioritized for those in need, as increased trips from other non-drivers could contribute to congestion.



THE UNIVERSITY OF
SYDNEY



Contact

Email: arkar.thanwin@sydney.edu.au
LinkedIn: Arkar Than Win

